

Week 6 - Advanced Analytics & Decision Support

No-Show Risk Analysis | Data Analytics Track

EXECUTIVE TAKEAWAY

Longer booking lead times and previous no-show history show the clearest observed association with non-attendance in the analysed dataset. These findings support targeted confirmation, reminder and scheduling interventions, while remaining subject to validation before causal or predictive conclusions are made.

1. Week 5 → Week 6 Transition

Week 5 established the HealthConnect analytical foundation: data preparation, exploratory analysis, five core KPIs and an initial Power BI dashboard. The most important baseline finding was an overall no-show rate of 48.46%, representing 2,423 no-shows out of 5,000 appointments. Week 5 identified deeper no-show risk-factor analysis as the next priority.

Week 6 therefore moved beyond the original EDA by concentrating on category-level no-show rates, appointment history, booking lead time and reminder-channel patterns, then translating those findings into business actions and a more decision-oriented dashboard.

2. Week 6 Objective and Scope

The objective was to deepen and validate the most decision-relevant Week 5 findings, improve the dashboard, identify evidence-based actions, document limitations and provide analytical inputs that can support the wider HealthConnect project.

Week 6 focus	Completed output
Advanced analysis	No-show rates by appointment type, previous no-shows, booking lead-time bands and reminder channel.
Validated findings	Cross-checking rates against the 5,000-record population and refined category-level comparisons.
Refined KPI	Overall no-show rate retained at 48.46% (2,423 / 5,000).
Dashboard improvement	Updated Week 6 Power BI analysis page with four decision-focused visuals and business insight/action panels.
Business relevance	Specific operational actions linked to the strongest observed patterns.

3. Data and Analytical Approach

The analysis continued using the HealthConnect appointment dataset containing 5,000 appointment records. The Week 6 approach used no-show rate rather than appointment counts alone, because category volume by itself does not establish higher risk.

- Calculated no-show rates across relevant categorical segments.
- Compared rates for patients with different previous no-show histories.
- Grouped booking lead time into interpretable bands: 0-2, 3-7, 8-14, 15-30 and 31+ days.
- Compared observed no-show rates across reminder channels.
- Reviewed small-volume categories cautiously and avoided treating them as reliable trend evidence.
- Updated the Power BI dashboard to communicate findings and recommended actions.

4. Key Analytical Findings

4.1 Overall No-Show Position

The overall no-show rate is 48.46%, with 2,423 no-shows among 5,000 appointments. This remains the core validated KPI and indicates that missed appointments are a material operational issue for HealthConnect.

4.2 Appointment Type

No-show rates across the four appointment types are relatively close, indicating that appointment type alone does not appear to be a strong differentiator in this dataset. The implication is that HealthConnect should avoid relying on appointment type as the sole targeting variable and should consider it alongside stronger behavioral and scheduling indicators.

4.3 Previous No-Shows

No-show rates increase consistently as previous no-show history increases across the main interpretable groups: 0 previous no-shows = 43.51%; 1 = approximately 53.49%; 2 = approximately 59.36%; and 3 = 67.95%.

BUSINESS IMPLICATION

Previous no-show history is one of the clearest observed risk signals. Patients with repeated prior no-shows may benefit from stronger confirmation and rescheduling processes.

4.4 Booking Lead Time

No-show rates rise steadily as booking lead time increases: 24.18% for 0-2 days, 30.05% for 3-7 days, 33.55% for 8-14 days, 43.21% for 15-30 days and 60.49% for 31+ days. The difference between the shortest and longest bands is 36.31 percentage points.

BUSINESS IMPLICATION

Long booking lead times are a major observed risk signal. Earlier confirmation, reminder scheduling and re-confirmation closer to the appointment date should be considered for appointments booked far in advance.

4.5 Reminder Channel

Observed no-show rates by reminder channel were: None 51.39%, WhatsApp 49.77%, Email 48.41% and SMS 45.75%. SMS has the lowest observed rate in this dataset, but the differences are observational and do not establish that SMS caused the lower no-show rate.

BUSINESS IMPLICATION

SMS is a candidate for further testing as part of a targeted reminder strategy. Its apparent advantage should be validated using controlled or appropriately adjusted comparisons before broad implementation.

5. Business Recommendations

Priorities long-lead appointments - Flag appointments booked 31+ days in advance for earlier and repeated confirmation activity.

Target repeated no-shows - Consider a more proactive confirmation/rescheduling workflow for patients with prior no-show history.

Test reminder effectiveness - Evaluate SMS against other reminder approaches using a structured test rather than assuming causality from the observed rates.

Use multi-factor targeting - Combine booking lead time and previous no-show history rather than targeting appointment type alone.

6. Dashboard Improvement

The Week 6 Power BI dashboard was redesigned from the Week 5 exploratory presentation into a more decision-oriented analysis page. The final page contains four equal-sized visuals, consistent muted-blue chart colors, a validated KPI, a Key Takeaway statement, and Business Insight/Action panels. The visuals focus specifically on the strongest no-show relationships identified during Week 6.

7. Data Limitations, Assumptions and Interpretation

- The analysis identifies associations and patterns; it does not establish causation.
- Reminder-channel comparisons may be affected by patient characteristics, operational processes or targeting decisions.
- Very small groups require caution. The 4-previous-no-shows and 5-previous-no-shows groups contain only 12 and 3 appointments respectively and were not used as primary evidence for the main trend.
- Category-level no-show rates should be interpreted together with sample size and, where possible, confidence intervals or statistical tests.
- The current analysis does not control potentially interacting variables such as patient characteristics, appointment timing and other operational factors.
- Findings should be validated on additional data and tested prospectively before being converted into operational rules.

8. Cross-Track Analytical Contribution

The Data Analytics contribution is intended to support the Data Science track by identifying candidate variables and relationships for further modelling. The Week 6 findings indicate that previous no-shows and booking lead time are particularly relevant candidate predictors, while reminder channel may warrant further investigation.

Collaboration status at submission: coordination/collaboration was attempted, but no response or model output was received from the other track before submission. Accordingly, this report does not claim that a Data Science model was changed as a result of the analysis. The updated Power BI dashboard and this documented analysis provide the evidence of the analytical contribution available at submission.

9. Week 7 Analytical Testing Requirements

- Test whether the booking lead-time relationship remains stable on new or additional data.
- Test whether previous no-show history improves prediction or risk stratification beyond other available variables.
- Evaluate reminder-channel effectiveness using appropriate statistical or experimental testing.
- Investigate whether patient and appointment characteristics confound the observed reminder-channel relationships.
- Apply minimum sample-size rules before interpreting high-risk categories.
- Compare Week 6 analytical findings with any Data Science model results and investigate material differences.
- Validate proposed interventions before wider operational implementation.

10. Week 6 Project Summary

Project Summary Area	Week 6 Position
Planned accomplishment	Deepen Week 5 no-show analysis, validate important findings, improve the dashboard and translate findings into business actions.
Completed	Four advanced no-show analyses, refined KPI presentation, redesigned Week 6 dashboard and documented business recommendations and limitations.
Improved from Week 5	Moved from broad EDA to rate-based segment analysis and decision-focused interpretation.
Integrated	Analytical findings were structured as candidate inputs for the Data Science modelling stage.
Collaboration	No collaborator response was received before submission; no unverified cross-track changes are claimed.
Key outcome	Booking lead time and previous no-show history emerged as the clearest observed risk indicators.
Challenge	Limited cross-track response reduced the ability to evidence a completed analytical hand-off before submission.

Important decision

Small-volume prior-no-show groups were treated cautiously rather than allowing extreme percentages to drive the conclusion.

Remaining issue

Relationships need further validation and, where appropriate, controlled or statistical testing.

Week 7 focus

Validate stability, test interventions, assess confounding and connect analytical findings with predictive modelling.